



Designing, developing, and deploying artificial intelligence systems: Lessons from and for the public sector

Kevin C. Desouza ^{a,*}, Gregory S. Dawson ^b, Daniel Chenok ^c

^a QUT Business School, Queensland University of Technology, 2 George Street, Brisbane City, QLD 4000, Australia

^b W.P. Carey School of Business, Arizona State University, Tempe, AZ 85287, U.S.A.

^c IBM Center for the Business of Government, IBM, U.S.A.

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Abstract Artificial intelligence applications in cognitive computing systems can be found in organizations across every market, including chatbots that help customers navigate websites, predictive analytics systems used for fraud detection, and augmented decision-support systems for knowledge workers. In this article, we share reflections and insights from our experience with AI projects in the public sector that can add value to any organization. We organized our findings into four thematic domains—(1) data, (2) technology, (3) organizational, and (4) environmental—and examine them relative to the phases of AI. We conclude with best practices for capturing value with cognitive computing systems.

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1. AI's history and value capture potential

Research and development of artificial intelligence (AI) systems has a rich history. There was a flurry of interest when AI was originally conceptualized in the 1950s, but it quickly fizzled in the face of AI's technical realities. Advances in

information and computational sciences over the last decade have provided the resources necessary to finally capitalize on the early discoveries and technology underpinning AI systems. As noted by Kai-Fu Lee (2018, p. 12), our current innovations in AI are “merely the application of the past decade’s breakthrough—primarily deep learning but also complementary technologies like reinforcement learning and transfer learning—to new problems.” Today, AI applications can be found in organizations across every sector and include chatbots that help customers navigate websites, predictive analytics systems used

* Corresponding author

E-mail addresses: kevin.c.desouza@gmail.com (K.C. Desouza), gregorysdawson@gmail.com (G.S. Dawson), chenokd@us.ibm.com (D. Chenok)

for fraud detection, augmented decision-support systems for knowledge workers, and semi- and fully autonomous systems in transportation, defense, and healthcare.

For the last 6 years, we studied AI systems across public, private, and nonprofit sectors. Our projects spanned multiple industry sectors, including healthcare, law enforcement, education, social services, defense, finance, management consulting, and infrastructure engineering. During these engagements, we had a firsthand view of critical issues organizations must contend with as they design, develop, deploy, and assess AI systems.

While most AI projects are complex, we assert that those within the public sector present some unique challenges:

- The public sector must contend with complex policy, societal, legal, and economic elements that might be skirted by their private-sector counterparts;
- Public-sector AI projects must advance the public good (Cath, Wachter, Mittelstadt, Taddeo, & Floridi, 2018) yet also deliver public value (Crawford, 2016a);
- These projects must go beyond simple cost and efficiency gains to satisfy a richer and diverse set of stakeholders who may have conflicting agendas;
- The need for transparency (Bryson & Winfield, 2017; Edwards & Veale, 2017) and fairness (Chouldechova, 2017; Crawford, 2016b) in decision making and system operations adds to the complexity of public-sector AI projects; and
- Given that public-sector projects and systems are taxpayer-funded, these efforts face regular scrutiny and oversight that is generally not seen in the private sector (BBC News, 2019).

While the aggregation of these factors is unique to the public sector, individual components apply to virtually every domain. For example, a nonprofit may have a much more diverse group of stakeholders than a privately held business, but a privately held business may have a higher requirement for cost and efficiency considerations. Thus, we believe that an examination of AI in the public sector can provide insights applicable to all organizations.

In this article, we share reflections and insights from our experience with AI projects in the

public sector. Within the phases of AI system engineering and implementation, we organized our findings into four thematic domains—(1) data, (2) technology, (3) organizational, and (4) environmental—and examine these domains relative to the phases of AI.

2. Background on cognitive computing systems

AI systems are often considered part of a larger set of *cognitive computing systems* (CCSs; Desouza, 2018). CCSs, as the name implies, have cognition due to their learning functions. A CCS can learn in one of two basic ways: supervised and unsupervised. In supervised learning, each record in the training dataset is tagged with its correct classification so that the machine learns what makes a record more or less likely to be in each group. In a fraud-detection exercise, each record is tagged as either fraudulent or nonfraudulent, and the machine identifies other attributes in the record that help to distinguish the two groups. The system may find that individuals from a certain school (e.g., University of Arizona) are more likely to commit fraud than those individuals from another school (e.g., Arizona State University). The system looks for the best distinguishing attributes to describe one group versus another.

In unsupervised learning, the system discovers previously unknown patterns or groupings in the data. This is common when a state looks to better understand its citizens. In this case, the machine runs, unsupervised, to discover patterns and types of people that exist within the state. A state may discover four different types of citizens along with their prevalence (e.g., few retirees and lots of families with young children). A human interprets the groupings that emerge; using this information, the state can make better-informed decisions about whether to fund more education or more services for seniors. Choosing between supervised and unsupervised learning is an exercise in understanding the trade-offs that exist between the accuracy of the learning and its interpretability. Thus, numerous selection considerations come into play when making the final decision (Lee & Shin, 2020).

Several characteristics differentiate CCSs from other systems. These five characteristics drive development and deployment choices:

1. CCSs learn from both data and human interactions, and both are required for successful deployment;

2. CCSs are context-sensitive and draw on environmental characteristics (e.g., user profiles and previous interactions) to deal with information requests;
3. CCSs recall history (i.e., previous interactions) in developing recommendations and groupings;
4. CCSs interact with humans through natural language processing; and
5. CCSs provide confidence-weighted recommendations (i.e., outcomes) that can be acted upon by humans.

Numerous examples of successful CCS deployments exist in the public sector (Desouza, 2018; Mehr, 2017). The U.S. government has implemented an AI-based chatbot app that helps potential refugees to the U.S. answer a series of questions to help determine required forms as well as assess whether a refugee is eligible for protection. North Carolina uses AI-based chatbots to answer basic help-center questions, which makes up approximately 90% of its phone calls. The Mexican government is piloting an AI tool to classify citizen petitions and then route them to the correct department. Finally, the organizers of the Pyongyang Winter Olympics developed and deployed an AI-based tool for real-time translation.

However, there have been some notable AI failures. Police departments have purchased an AI-based tool that executes real-time facial recognition. Unfortunately, the tool returned a high number of false positives and falsely matched 28 members of Congress with mugshots of unrelated individuals (Santamicone, 2019). The city of Chicago, in an effort to identify people most likely to be involved in a shooting, developed an algorithm to identify such individuals and stop them from making a firearm purchase. Yet, a report by the RAND Corporation showed that the output from the tool is less effective than a most-wanted list and, even worse, targets innocent citizens for police attention (Stroud, 2016).

These examples show the power of CCS deployment in the public sector as well as the potential risks. It is not surprising that technology experts generally agree on the power of AI to transform the economy and society but remain sharply divided over whether the transformation will be helpful or harmful (Kietzmann & Pitt, 2020). Managers are responsible for deciding if, where, why, and how AI should be adopted so as to capture its benefits while mitigating its risks (Kietzmann & Pitt, 2020).

3. Designing cognitive computing systems

Organizations need to be strategic when designing CCSs and in choosing a low-hanging-fruit initiative and a bold challenge. Then, they need to ensure the proper infrastructure is in place, both internally and externally, to complete the project successfully.

3.1. Low-hanging fruit vs. grand challenge

With a low-hanging-fruit initiative, the organization might look at a problem for which automation through the deployment of a CCS can alleviate mundane work and increase efficiencies. In a grand challenge, the organization can look at how CCSs open opportunities for business-model transformation and disruptive innovation. Indeed, the use of CCSs can play a key role in developing innovation through the judicious application of analytics (Kakatkar, Bilgram, & Füller, 2020).

Organizations with immature information-systems capability often benefit from tackling low-hanging fruit first—a wise strategy for all types of systems development but even more appropriate for a CCS. This allows them the opportunity to gain experience working through the various phases of system development and deployment. They can build on existing data assets and system capabilities while extending their current knowledge base and expertise. However, organizations that have deep in-house expertise in information-system development and/or have sufficient IT resources can successfully engage in a bold challenge. Here, the organization can leverage the collective wisdom to identify key opportunities for CCS design that align with internal or environmental disruptions. As part of this initiative, the organization can launch an educational campaign to increase the workforce's familiarity with cognitive-systems techniques and applications. This does not mean that the organization should only look internally for necessary resources. All organizations, but particularly public-sector organizations, have a rich ecosystem of commercial partners that can contribute to the effort.

Technologically sophisticated organizations ready to approach a bold challenge are the exception rather than the rule. While these advanced firms infuse innovation into their corporate DNA, most organizations lack the maturity necessary to undertake a major cognitive computing effort without first testing the waters and improving with easier, lower-risk projects.

Finding small success with these low-hanging fruits will help the organization to mature and build some credibility before tackling a bold challenge.

3.2. Ensure necessary capabilities

At the start, organizations must take a close look at data, technological, organizational, and environmental elements to determine if any CCS implementation is feasible and, if so, which type will most likely yield a successful result. Indeed, care must be taken to see that the proposed solution does benefit the organization since poorly planned or conceived CCSs can hinder rather than help an organization's value chain (Canhoto & Clear, 2020).

3.2.1. Data

The organization must determine whether the data are available, accessible, and analyzable in order to take advantage of CCS algorithms. These issues need to be considered carefully if the organization is tackling an accessible opportunity. While there will be costs associated with readying data due to cleaning and connecting (i.e., integrating), they should not be exorbitant. However, if the data are not available, it could be a sizable and costly effort to find, validate, and incorporate the data into the CCS scheme. Absent available data, the effort may be classified as a grand challenge.

In the case of grand challenges, ideas should be solicited regarding what data to analyze, what data sources need to be procured and integrated, and what major trials the organization will confront as data is readied. Within the public sector, processes have evolved to deal with properly communicating the data risks of these grand challenges before the effort begins. Unfortunately, the public sector has perhaps over-engineered these processes, which could explain why they take so much time.

If a private-sector organization goes after a grand challenge, it needs to develop and implement a repeatable process for properly understanding the advantages and disadvantages of collecting such data. In addition to the simple technical aspects of collecting, cleansing, and deploying the data, legal and ethical implications must be considered. The stakes with these projects are high, and it is often helpful to collaborate with outside experts to steer the data procurement process, although they should have no direct or indirect role in the implementation to avoid conflicting goals.

3.2.2. Technology

On the technological front, the organization must have a good handle on its current IT assets from both infrastructure and capability perspectives. Regarding infrastructure, does the organization have the necessary IT applications and allied technical resources to undertake a CCS development effort? If not, can the organization foster existing or new partnerships to access the necessary technical resources? Governments are particularly apt at identifying skilled partners to support the work. A private-sector organization can also do more to leverage other potential partners who can provide the missing resources and, ideally, share in the risk and the reward of the initiative. Some of the public agencies we studied use a risk-versus-value determination for aligning with partners. Risk can be high or low and value can be high or low (see Table 1), and the particular result can drive the partnership decision.

In the case of government, the most salient domain is risk; the natural tendency is to try to minimize risk rather than maximize value. In contrast, private-sector organizations are more likely to prioritize value over risk, and that can alter their approach. The organization needs to take a careful look at what expertise/knowledge should be developed internally versus what can be sourced.

As we will discuss later, CCSs require training from human experts and should be subject to audits; organizations should carefully consider how they continue to develop technical capabilities to (1) retain the expertise to design the next-generation of CCSs, and (2) investigate, fix, and learn from CCS deployments. If too much work is outsourced, the organization faces costly support for future deployments, and this may be particularly acute in private-sector organizations more focused on value than risk.

3.2.3. Organization

The organizational front looks internally to understand the organization's current strengths and weaknesses. For organizations new to CCSs, the

Table 1. Risk and value typology

	Low value	High value
Low risk	Explore partnership	Perform work in-house
High risk	Outsource work	Explore partnership

organizational front is considerably more challenging. Organizations often have trouble assessing their own capabilities in a candid way.

This organizational front problem is even more acute in government due to the long tenure of most government managers and their frequent lack of comparative outside experience. Without an external frame of reference, an appropriate assessment is more challenging. Agencies have made some progress in this area through data sharing, which helps all levels of government face similar challenges and limitations.

In contrast, very few people in the private sector have spent their careers with just one firm, and most managers have frames of reference against which to evaluate their current employers. If the organizational assessment reveals a lack of maturity, the organization can simply buy/hire or rent/contract with firms or individuals with the necessary skills.

3.2.4. Environment

The environment front looks at any efforts underway at other companies in the same or adjacent industries. The goal of the environment scan also differs for public-sector versus private-sector organizations due to the natures of the domains. A realization that other government entities use CCSs allows the public sector to request necessary information and then apply that to their problem. However, in addition to commercial companies being unwilling to share their confidential information, competition can drive CCS adoption in order to avoid being boxed out of the market by a rival firm already progressing strongly in this area. [Table 2](#) summarizes our findings for choosing a CCS initiative.

A government can perform an environment scan more easily since most civilian government applications are transparent. Thus, public-sector organizations can share this information to develop a clear sense of environmental status. Not surprisingly, the private sector has a bigger hurdle with this since most organizations will not freely share their competitive plans with others in or adjacent to their industry. Thus, the private sector needs to do a great deal more speculating on the maturity of efforts in other companies.

4. Developing cognitive computing systems

Best practices for developing a CCS are the same as those necessary for developing noncognitive systems: Organizations must uncover

requirements, follow a robust methodology, and involve skilled developers, project managers, and users to ensure successful development. In this section, we address some of the key differences in developing a CCS versus other systems.

CCSs learn from data, and thus the focus is on being able to understand the data and potential biases ([Wiggers, 2018](#)). [Wiggers \(2018\)](#) identified notable examples of these biases:

- An AI system was used to predict whether a defendant would commit a future crime. The algorithm was found to overstate the risk for black defendants versus white defendants.
- A study by Boston University found that datasets used to teach AI systems contained “sexist semantic connections” and, for example, considered “programmer” to be more masculine than feminine.
- A study by MIT found that facial recognition software was 12% more likely to misidentify black males than white males. The problem of bias is real and needs to be addressed in the development stage of the CCS.

New York City confronted this challenge with efforts to make algorithms more accountable ([Powles, 2017](#)). A 2018 law created a task force to examine the city’s CCSs—which guide allocation and deployment of resources ranging from police officers and firefighters to public housing and food stamps—to make sure data and support algorithms are fair and transparent. As James Vacca, a New York city council member, said: “If we’re going to be governed by machines and algorithms and data, well, they better be transparent” ([Powles, 2017](#)). An initial part of this legislation, since dropped, would have required a city agency to turn over the source code of any new CCS to the public and also to simulate the algorithm’s performance using data provided by New Yorkers.

Since systems will support decisions based on data, there is value in understanding the data and its biases; the data and its algorithms need to be checked and validated prior to deployment. In determining the quality of an algorithm, it is necessary to envision the risk dimension we display in [Table 1](#). For the public sector, the risk dimension remains much more salient; as the risk of making a bad decision rises from low to high, more evaluation of the data and its algorithms becomes necessary. For the private sector, a more balanced assessment of risk versus value drives decision making.

Table 2. CCS design issues

Category	CCS Challenge	Public-Sector Strategies	Private-Sector Strategies
Data	• Data availability • Data sources identification • Legal/ethical challenges	• Assess data availability, accessibility, and analyzability • Walk away from grand challenges if data not acceptable	• Develop a repeatable process for collecting data • Determine harm from poor decision making • Hire experts to steer the process
Technical	• Current asset identification • Identification of complementary partners • Risk versus value determination	• Focus on risk dimension • Internal expertise versus partnerships • Eye on future cost	• Balance risk and value • Internal expertise versus partnerships
Organizational	• Challenge in quality assessment	• Engage outside experts • Sharing data across government • Hire missing expertise	• Leverage network of internal staff
Environment	• Challenge in disclosure of information	• Leverage inherent government transparency	• Leverage network of internal staff • Engage especially if other firms are engaging

The technical dimension of deploying CCSs has improved over time due to the development of open-source tools for auditing algorithms. In 2018, a New York-based AI startup announced that it had decided to open source its audit tool, used “to determine whether a specific statistic or trait fed into an algorithm is being favored or disadvantaged at a statistically significant, systematic rate, leading to adverse impact on people underrepresented in the dataset” (Johnson, 2018). However, the tool does not actually remove the bias but merely points the bias out to the developer. The release of such open-source tools comes as the industry continues to create tools that can also detect biases (Wiggers, 2018).

Organizational issues in CCSs are probably the most challenging since they likely will not have people with the requisite skills in-house. In the public

sector, this has been partially solved with the use of agile acquisition practices. Agile acquisition is a procurement strategy that “integrates planning, design, development, and testing into an iterative lifecycle to deliver small, frequent, incremental capability to the end user....[it] requires integrated government and contractor processes and partnerships” (MITRE, 2019). The proposal process differs at the government level because it can require the bidders to complete a sprint rather than a written or oral proposal (White, Sateri, & Chawanasunthornpot, 2018). Although not yet adopted for CCS development by any known government entity, agile acquisition can dramatically lower the risk associated with development. Rather than attempting to acquire an entire system with a single procurement—a risky and lengthy endeavor—the government can buy

Table 3. CCS development issues

Category	CCS Challenge	Public-Sector Strategies	Private-Sector Strategies
Data	• Inadvertent bias in data and algorithms	• Risk-focus determination on how good data and algorithms need to be	• Balanced risk/value determination on how good data and algorithms need to be
Technical	• Availability of tools to audit for bias	• Take advantage of tools	• Take advantage of tools
Organizational	• Lack of trained staff	• Agile acquisition strategy	• Agile acquisition strategy
Environment	• Availability of tools to audit for bias	• Take advantage of tools	• Take advantage of tools

individual pieces of the system. Given the newness of CCS development in government, this should dramatically lower the risk.

Private-sector firms can adopt agile acquisition for the development of their CCSs since the same-risk reduction goals exist as with the public sector, and the use of this strategy should build skill levels available to the organization. Environmental issues closely track to technical issues during the development phase. The right tools can aid in the development of the CCS, many of which can tackle biases in the data. Given the expansion of open-source tools, the adoption of such approaches should increase significantly. Table 3 summarizes our CCS development findings.

5. Deploying cognitive computing systems

While a great deal of work goes into CCS deployment, a significant amount of work needs to happen after deployment. Two aspects are particularly salient. First, these systems must be audited. While all systems need monitoring, continuously learning CCSs can learn from the wrong data. Algorithmic biases can easily sneak into a CCS, and the organization needs to understand the risk and preemptively test for it. In 2016, the Obama administration directly called on companies to audit their algorithms. DJ Patil, former chief data scientist for the U.S., said at the time: “Right now the ability of algorithm to be evaluated is an open question. We don’t know technically how to take the tech box and verify it” (Hempel, 2018). Unfortunately, there are few standards for auditing an algorithm; even if such criteria existed, a company may have to disclose trade secrets to the auditor. Singapore and Korea recently announced the formation of an AI ethics board, as have various industry groups.

An audit could have value by revealing that a company does not discriminate against different

classes of individuals. By passing an audit, the company could use marketing success as a sort of “Good Housekeeping seal of approval, signaling to customers a higher standard of ethics and building trust” (Hempel, 2018). In New York, a rich market is developing ways to examine and certify algorithms. In the New York City rental market, one company has developed a program to grade the maintenance of every rental building in the U.S. Using information from 311 phone calls to the city—which lodge violation complaints against the building—this program calculates the grades and will certify it for an additional fee of \$100–\$1,000 per year. An external examiner evaluated the program’s logic and found that the algorithm was fair (Hempel, 2018).

Second, the value of the CCS needs to be determined. This taps the external dimension. The public sector must consider value more holistically than simply efficiency gains. The concept of public value (Osborne, Radnor, & Nasi, 2013) has a rich history in the public management literature. Common elements captured in various public values frameworks include equity, fairness, promotion of the common good and public interest, protecting the vulnerable and human dignity, economic sustainability of the program/initiative, and upholding the rule of law (e.g., Benington & Moore, 2011; Chohan & Jacobs, 2018; Jørgensen & Bozeman, 2007).

Organizations should examine efforts in the public sector to consider more holistic approaches to capturing value from CCSs. We recommend thinking of value capture along four dimensions, and the output of this thinking can be reflected in the value/risk typology we presented in Table 1. The following values can be derived:

- Process gains: costs or resources saved, time cut from processes, etc.;
- Output gains: being more effective;

Table 4. CCS deployment issues

Category	CCS Challenge	Public-Sector Strategies	Private-Sector Strategies
Data	Inadvertent bias in data and algorithms	Audit of the algorithm to ensure accuracy	Audit of the algorithm to ensure accuracy
Technical	Availability of tools to audit for bias	Take advantage of tools	Take advantage of tools
Organizational	Lack of trained staff	Take advantage of tools	Take advantage of tools
Environment	Availability of tools to audit for bias	Take advantage of tools	Take advantage of tools

- Outcome gains: transforming the customer experience and acting ethically; and
- Network gains: shaping interactions with other stakeholders due to the CCSs, adding to brand and leadership position, capturing new value from an existing supply network, and adding value through new affordances and influences to the overall ecosystem through disruption and transformation.

The first two address the organization, ensuring that it 'advances the business of today' through CCSs, while the last two ensure that it focuses on 'advancing the business of tomorrow' and being relevant in the age of CCSs. We summarize the issues in Table 4.

6. Harnessing new technologies

CCSs will change the nature of computing in a dramatic way, and we are seeing the benefits of their implementation in all aspects of government and private industry. We are entering what futurists already call the Fourth Industrial Revolution (Schwab, 2017). In this fourth revolution, technologies allow a convergence of computing power as transformative as the previous revolutions, but with much greater reach.

On the positive side, this means the world may become more connected than ever before, dramatically improving the efficiency of organizations and potentially even regenerating the damage of previous revolutions. However, as pointed out by Klaus Schwab, this requires organizations to adapt in novel and challenging ways and governments to manage new technologies and address security concerns (Schwab, 2017). Schwab calls for leaders and citizens to work together to harness these new technologies. We concur with this call and recommend a necessary first step to acknowledge that technologies like CCSs introduce challenges that need new management

approaches, like those outlined in this and other articles in this special issue.

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